MIL-I-7723B

1 June 1965
SUPERSEDING
MIL-I-7723A
7 April 1952

MILITARY SPECIFICATION

INDICATOR, TEMPERATURE, ELECTRICAL RESISTANCE, MULTIFUNCTION, -70° TO ±150° C, DUAL

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope - This specification covers one type of dual pointer, electrical resistance type temperature indicator.

1.2 Classification - The indicator shall be classified as shown on MS90375.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on the date of invitation for bids, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-P-116	Preservation, Methods of
MIL-N-3336	Nut, Self Locking, Instrument Mounting
MIL-C-5015	Connectors, Electric, AN Type
MIL-I-7070	Indicator, Temperature, Electrical Resistance, Hermetically Sealed, General Specification for
MIL-P-7936	Parts and Equipment Aeronautical, Preparation for Delivery
MIL-D-70327	Drawings, Engineering and Associated Lists

STANDARDS

Federal

FED-STD-595	Colors

FSC 6620

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Military

MIL-STD-130	Identification Marking of U. S. Military Property
MS90375	Indicator, Temperature, Electrical Resistance Multifunction, -70° to +150° C, Dual

(Copies of the applicable documents required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 General - The requirements specified in MIL-I-7070 are applicable as requirements of this specification. Additional requirements shall be as specified herein.

3.1.1 In the event the requirements of the general specification and this specification conflict, the requirements of this specification shall govern.

3.2 Construction -

3.2.1 Case - The case shall be as specified in MIL-I-7070 except that it shall be finished with a lusterless black material, Color No. 37038 of FED-STD-595 and shall conform to MS90375.

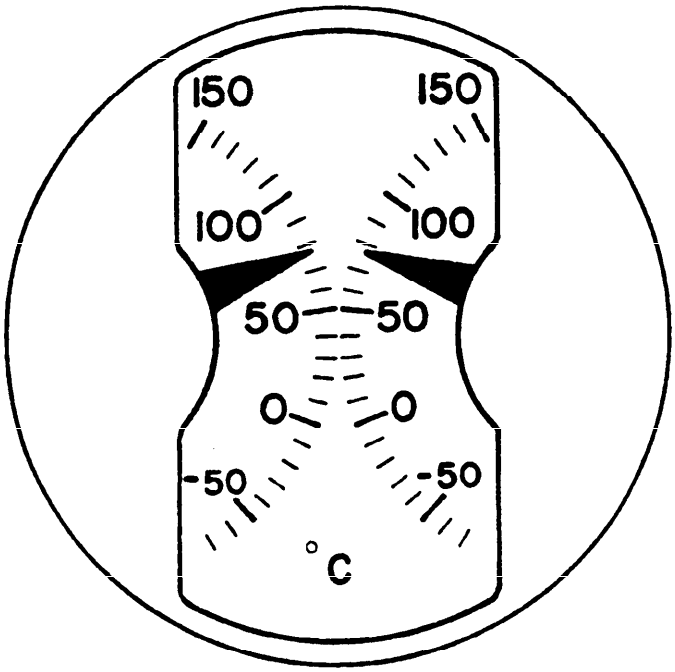
3.2.2 Scale length - The scale of the indicator shall be as long as practicable and shall be not less than 1-3/4 inches as measured on the outer arc of the shortest graduations.

3.2.3 Pointers - The pointers shall be as light as practicable and sufficiently rigid to prevent oscillation under vibration. Pointer stops, if used, shall be so located as to permit the pointer to move not less than 5°C beyond the end of the scale. The pointer length shall be such that the pointer tip will extend into the scale to a distance equal to 1/4 to 1/2 of the length of the shortest graduation. The pointers shall not touch each other at midscale. The minimum length of pointer showing shall be 1/2 inch. The width of the pointer 1/2 inch from the tip shall be 5/64 ±1/64 inch and shall taper to a point, or the width of the pointer 5/16 inch from the tip shall be 5/64 ± 1/64 inch and shall taper to a point.

3.2.4 Dial - The dial shall conform to Figure 1.

3.2.5 Electric connector - The electric connector used shall conform to MIL-C-5015 and MS90375. The connector shall be wired as follows:

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MARKING	HEIGHT OR LENGTH ±0.010	WIDTH OF LINE ±0.005	MATERIAL OR FINISH
NUMERALS -50	0.156	0.030	LUSTERLESS WHITE COLOR NO. 37875 OF FED-STD-595
NUMERALS 0, 50, 100, AND 150	0.187	0.030	
VISIBLE PORTION OF POINTERS	- - -	- - -	
MAJOR GRADUATIONS	0.125	0.020	
MINOR GRADUATIONS	0.094	0.010	
LETTERING °C	0.187	0.030	
BACKGROUND OF DIAL	- - -	- - -	LUSTERLESS BLACK COLOR NO. 37038 OF FED-STD-595

DIMENSIONS IN INCHES

FIGURE 1. Dial, Pointers, and Markings

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Connector Pin

A	+ Battery
B	Resistance bulb (left)
C	Common-Negative battery and remaining resistance bulb connection
D	Open
E	Resistance bulb (right)

3.2.6 Weight - The weight of the complete indicator shall not exceed 1.6 pounds.

3.3 Identification of product -

3.3.1 Nameplate - A nameplate shall be securely attached to the exterior of the case and shall be marked in accordance with the requirements of MIL-STD-130 except that the FSN shall be omitted.

3.3.2 Manufacturer's part number - The manufacturer's part number on the nameplate shall be identical with the manufacturer's engineering production drawing number including applicable dash numbers if the drawing is tabulated and covers more than one part.

3.4 Interchangeability - All parts having the same manufacturer's part number shall be directly and completely interchangeable with each other with respect to installation and performance. Changes in manufacturer's part numbers shall be governed by the drawing number requirements of MIL-D-70327.

3.5 Mounting hardware - The contractor shall furnish mounting screws and self locking nuts conforming to MIL-N-3336 in sufficient quantity to install the indicator. The screws shall be round-head brass machine screws having a durable, lusterless black, oxidized or nickel finish. The length shall be sufficient to mount the indicator on panels up to 3/16 inch thick.

3.5.1 Envelope - An envelope furnished by the contractor containing mounting screws and nuts shall be packaged with each instrument and shall be marked with the following information:

"IMPORTANT
THIS ENVELOPE CONTAINS
MOUNTING SCREWS AND NUTS"

3.6 No data is required by this specification (other than reports accompanying samples submitted for qualification testing), or by applicable documents referenced in Section 2, unless specified in the contract or order (See paragraph 6.2).

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 General - Sampling, inspection, and testing of the indicators shall be in accordance with MIL-I-7070 and as specified herein.

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Inspection conditions -

.2.1 Test voltage - The test voltage shall be 28.5 ±0.5V DC, otherwise specified.

.3 Tolerances - The tolerances specified in Table I, not en- in parentheses, are applicable only when the indicators are subjected the tests in the sequence specified herein. The tolerance specified le I, enclosed in parentheses, are applicable only when the indicators bjected to a specific test independent of sequence.

.4 Inspection methods - The examination of product test of 7070, the tests listed in Table I, and the following apply. Unless ise specified, tests shall be conducted as described in MIL-I-7070.

.4.1 Position error - When the indicator is subjected to the on error test, the maximum change in indication shall not exceed 4°C.

.4.2 Friction error - When the indicator is subjected to the on test, the maximum difference in reading before and after tapping not exceed 4°C.

.4.3 Insulation leakage - When the indicator is subjected to sulation leakage test, the insulation resistance shall exceed 20 megohms.

.4.4 Reduced voltage operation - The indicator shall be sub- to a Scale error at room temperature test with an applied voltage of + the test points specified in Table I. The indications at the re- tage shall not differ from the indications with 28.5V applied by an the tolerances specified in Table I.

.4.5 Magnetic effect - When the indicator is subjected to the ic effect test, it shall not cause the magnetic compass to deflect more degrees.

.4.6 Resistance bulb current - A total resistance of 100 ohms be applied across the bulb circuit terminals of the indicator, the oltage applied, and the bulb circuit current measured. The current con- in the resistance bulb circuit shall not exceed 22 ma.

.4.7 Vibration error - When the indicator is subjected to the ion error test, the maximum amplitude of pointer oscillation through- e frequency range shall not exceed 4°C. The maximum difference between iginal position of the pointer before vibration and the position of the r during vibration shall not exceed 4°C. Pointer oscillation shall not sidered as pointer variation.

.4.8 Response time - When the indicator is subjected to the se time test, the response time shall not exceed 3 seconds.

.4.9 Damping factor - When the indicator is subjected to the g factor test, the damping factor shall be not less than 1.6.

.4.10 Power consumption - When the indicator is subjected to the

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TABLE I - Test Tolerance

Test Point	Test Resistance	Scale Error at Room Temperature	Reduced Voltage	Scale Error at Low Temperature	Scale Error at High Temperature	Scale Error at Room Temperature After Open and Short Circuit
°C	Ohms	Tolerance ± °C	Tolerance ± °C	Tolerance ± °C	Tolerance ± °C	Tolerance ± °C
-70	68.27	3.0 (3.0)	2.0 (2.0)	7.0 (7.0)	7.0 (7.0)	10.0 (4.0)
-30	80.56	3.0 (3.0)	2.0 (2.0)	6.0 (6.0)	6.0 (6.0)	10.0 (4.0)
0	90.38	2.0 (2.0)	2.0 (2.0)	4.0 (4.0)	4.0 (4.0)	10.0 (3.0)
+30	100.91	3.0 (3.0)	2.0 (2.0)	6.0 (6.0)	6.0 (6.0)	10.0 (4.0)
+80	120.36	3.0 (3.0)	2.0 (2.0)	6.0 (6.0)	6.0 (6.0)	10.0 (4.0)
+120	137.78	3.0 (3.0)	3.0 (3.0)	6.0 (6.0)	6.0 (6.0)	10.0 (4.0)
+150	151.91	3.0 (3.0)	3.0 (3.0)	7.0 (7.0)	7.0 (7.0)	10.0 (4.0)

Test Point	Test Resistance	Scale Error at Room Temperature After Vibration Failure	Extreme Low Temperature Exposure		Scale Error at Room Temperature After High Temperature Exposure
			Scale Error at Low Temperature	Scale Error at Room Temperature After Low Temperature Exposure	
°C	Ohms	Tolerance ± °C	Tolerance ± °C	Tolerance ± °C	Tolerance ± °C
-70	68.27	10.0 (7.0)	14.0 (7.0)	10.0 (3.0)	10.0 (3.0)
-30	80.56	10.0 (7.0)	13.0 (6.0)	10.0 (3.0)	10.0 (3.0)
0	90.38	10.0 (6.0)	12.0 (4.0)	10.0 (2.0)	10.0 (2.0)
+30	100.91	10.0 (7.0)	13.0 (6.0)	10.0 (3.0)	10.0 (3.0)
+80	120.36	10.0 (7.0)	13.0 (6.0)	10.0 (3.0)	10.0 (3.0)
+120	137.78	10.0 (7.0)	13.0 (6.0)	10.0 (3.0)	10.0 (3.0)
+150	151.91	10.0 (7.0)	14.0 (7.0)	10.0 (3.0)	10.0 (3.0)

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ɔnsumption test, the total power consumption shall not exceed 2.0

4.11 Bulb lead resistance effect - The indicator shall be con-
as shown in Figure 2. A scale error test at room temperature shall
ɔrmed for each indicator movement, with and without the 2 ohm resis-
:ach lead. The change in indication due to the added resistance
ɔt be greater than 3°C.

EPARATION FOR DELIVERY

Packaging - The indicator shall be packaged in accordance
-P-7936, Level A or C as specified in contract or order (See 6.2).
el A packaging the method of preservation shall be in accordance
-P-116, Method 1A-5, without preservative compound.

Packing - The indicator shall be packed in accordance with
936, Level A, B, or C as specified in contract or order (See 6.2).

Marking - The interior and exterior containers shall be
as specified in MIL-P-7936.

3.1 Precautionary marking - The following precautionary mark-
ill appear on two opposite sides of each interior package and ship-
itainer wherever practicable.

FRAGILE
DELICATE INSTRUMENTS
HANDLE WITH CARE

TES

Intended use - The electrical resistance temperature indi-
ɔvered by this specification is intended to indicate temperatures in
ge -70° to +150°C, such as oil and coolant.

Ordering data - Procurement documents should specify the
ig in addition to ordering data listed in MIL-I-7070:

(a) Title, number and date of this specification.

(b) The quantity, MS part number including dash number and type number, if any, of the indicator desired.

(c) Data requirements.

General - The instructions contained in Section 6 of MIL-I-
:luding provisions for Qualification tests are applicable to and form
ɔf this specification except that information pertaining to qualifi-
ɔf products covered by this specification shall be obtained from:
Bureau of Naval Weapons, Department of the Navy, Washington, D. C.

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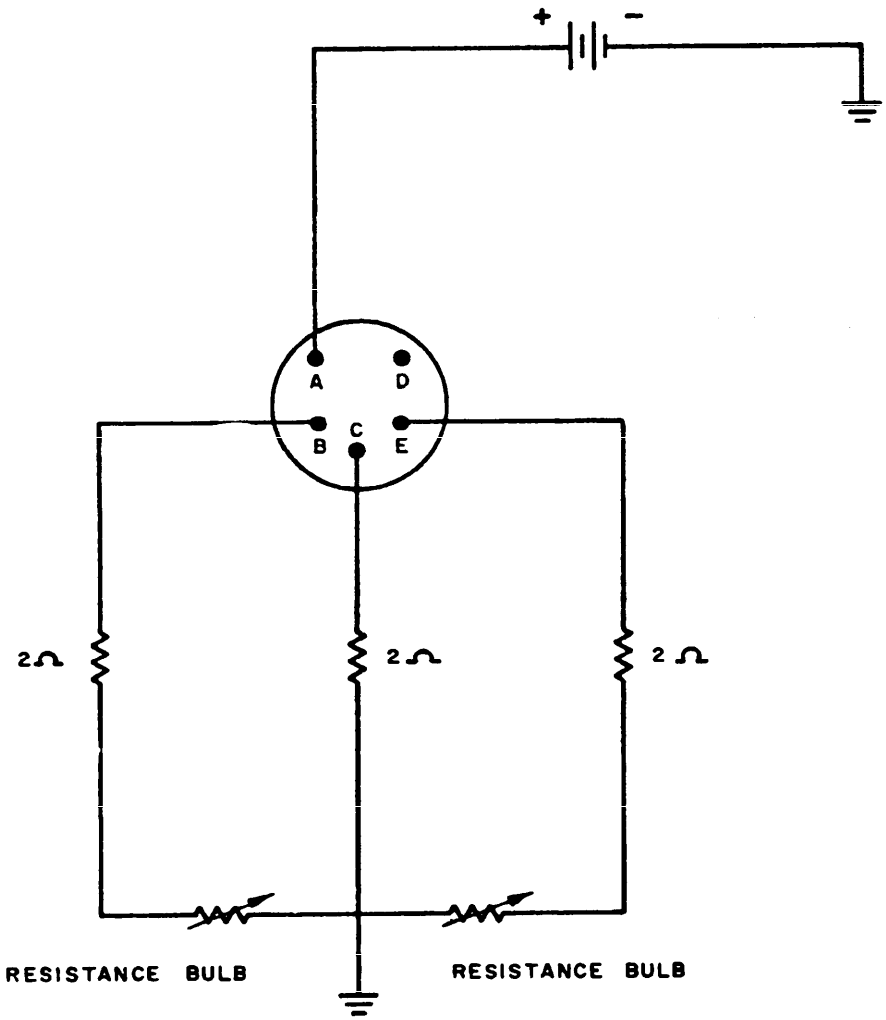


Figure 2 BULB LEAD RESISTANCE TEST

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6.4 Definitions -

6.4.1 "Inspection" as used in the specification includes examination and tests.

6.5 Changes from previous issue - The outside margins of this specification have been marked "#" to indicate where changes from the previous issue have been made. This has been done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content as written irrespective of the marginal notations and relationship to the last previous issue.

6.6 Precedence of documents - When the requirements of the contract, this specification, or applicable subsidiary specifications are in conflict, the following precedence shall apply:

(1) Contract - The contract shall have precedence over any specification.

(2) This specification - This specification shall have precedence over all applicable subsidiary specifications. Any deviation from this specification, or from subsidiary specifications where applicable, shall be specifically approved in writing by the procuring activity.

(3) Referenced specifications - Any referenced specification shall have precedence over all applicable subsidiary specifications referenced therein. All referenced specifications shall apply to the extent specified.

Custodians:
Air Force - No interest
Army - MO
Navy - WP

Review activities:
Air Force -
Army - MO
Navy - WP

User activities:
Air Force -
Army -
Navy - MC, CG

Preparing activity
Navy - WP

Proj. No. 6685-0108

Review/user information is current as of date of this document. For future coordination of changes to this document, draft circulation should be based on information in the current Federal Supply Classification Listing of DOD Standardization Documents.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION MIL-I-7723B INDICATOR, TEMPERATURE, ELECTRICAL RESISTANCE, MULTIFUNCTION, -70° TO +150°C, DUAL

ORGANIZATION *(Of submitter)* | CITY AND STATE

CONTRACT NO. | QUANTITY OF ITEMS PROCURED | DOLLAR AMOUNT

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED *(Printed or typed name and activity)* | DATE

DD FORM 1 NOV 64 1426 REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE C-8279

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Naval Air Engineering Center
Philadelphia, Pennsylvania 19112

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